



Powering up devices

An overview of various battery technologies being developed to power the electronics of tomorrow. <https://digit.in/may20-79>



Aiming for the atom

With computer components now being made atom by atom, here is how chips will continue shrinking. <https://digit.in/may20-80>

Materials for future electronics

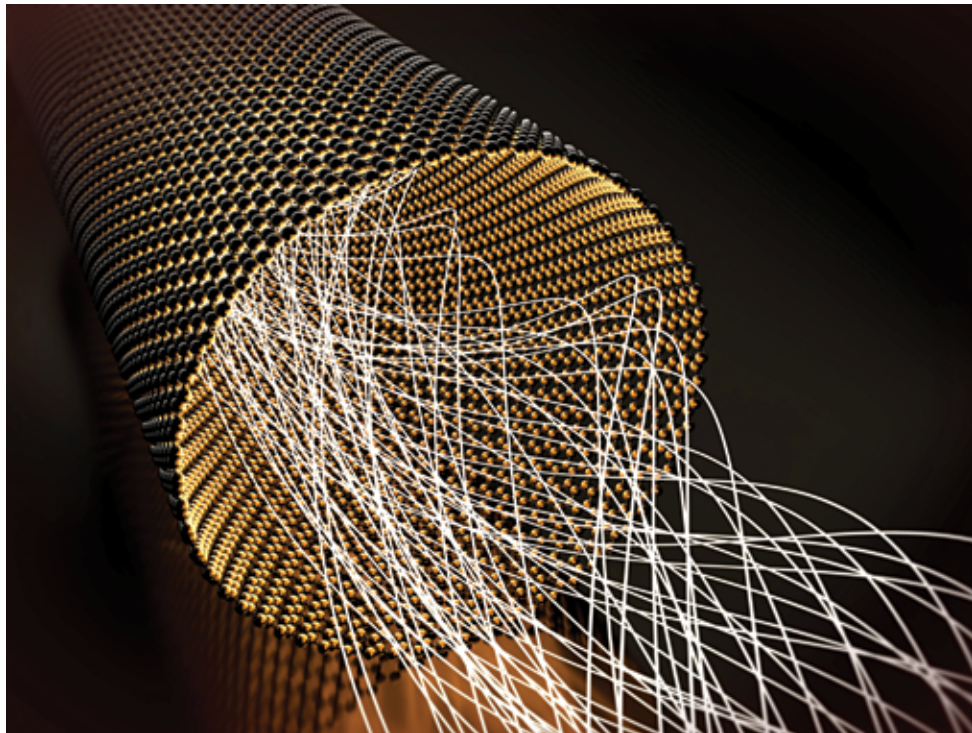
Nanoscale, energy efficient and even biodegradable, this is what the future of electronics looks like

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IoT and wearables require electronics to be everywhere, and this would require devices to be far

more flexible. Phones that bend are already here, or at least just around the corner. In the labs though, researchers are already making electronics from materials that mimic the properties of the human skin. Not only can these electronics flex and twist, they can also stretch, repair themselves and some of them are even biodegradable. The first example of semiconducting polymer that is biodegradable, requires the presence of a weak acid, such as vinegar, and has been developed by researchers at MIT. The approach allows sensors to be attached directly to the skin, and even on sensitive tissues such as the brain. While traditionally electronics that could be applied on the skin used gold components, researchers have now made components using cheap iron. Iron is environment-friendly and is not harmful to humans. The research allows for the creation of sensor dust, such as the ability to drop sensors over a forest area for monitoring purposes through airplanes, without damaging the environment. This is just one of the



many new types of electronics being studied in laboratories.

2D SHEETS

Electronic components are being built in labs that are put together atom by atom. These are known as 2D materials for sheets, and 1D materials for wires. In either case, the materials are just a single atom in thickness. One of the earliest 2D materials to be made was graphene, which allows for strong sheets of carbon to be made, even though they are a hundred times thinner than a razor.

When it comes to electronics, magnets are ubiquitous. Hard disk drives have some pretty powerful magnets, and so do speakers. Within the smartphone, the digital compass to tell the direction, the magnetometer to track the orientation of the smartphone

in three dimensions, as well as the speakers, all have magnets. Most of the storage used in modern electronics is based on the principle of magnetic switching, where switches are toggled to store data. However, the exact mechanism of achieving this is done by passing an electrical current through a coil, creating a magnetic field that activates the switches. This is demanding in terms of energy, and also results in waste heat.

While researchers were aware of the possibility of 2D magnets existing since 1966, it was only in 2017 that such magnets were actually created at Cornell University. The researchers used a material known as chromium triiodide to create magnets that were a single atom in thickness. The biggest advantage is that the compound's magnetism can be directly con-